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VEHICULAR EXTENDABLE CAMERA ASSEMBLY

Abstract

A vehicular camera assembly includes a mounting base configured to mount at a vehicle, an arm pivotally mounted at the mounting base, a camera disposed at an outer end region of the arm, and an actuator that is operable to pivot the arm between a drive position and a folded position. With the vehicular camera assembly mounted at the vehicle, and when the arm is pivoted from the drive position toward a forward fold position, the arm is at least partially received in a forward cavity at the side of the vehicle. A forward flap assembly seals the forward cavity when the arm is pivoted toward the forward fold position. The forward flap assembly, when the arm is pivoted away from the folded position and beyond the drive position toward the forward fold position, engages the arm and biases the arm toward the drive position.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application is a division of U.S. patent application Ser. No. 18/444,842, filed Feb. 19, 2024, now U.S. Pat. No. 12,311,845, which is a continuation of U.S. patent application Ser. No. 17/661,753, filed May 3, 2022, now U.S. Pat. No. 11,912,204, which claims the filing benefits of U.S. provisional application Ser. No. 63/202,633, filed Jun. 18, 2021, and U.S. provisional application Ser. No. 63/201,527, filed May 4, 2021, which are hereby incorporated herein by reference in their entireties. U.S. patent application Ser. No. 17/661,753 also is a continuation-in-part of International Application No. PCT/US2021/070765, filed Jun. 24, 2021, which claims the filing benefits of U.S. provisional application Ser. No. 63/202,633, filed Jun. 18, 2021, U.S. provisional application Ser. No. 62/706,640, filed Aug. 31, 2020, and U.S. provisional application Ser. No. 62/705,371, filed Jun. 24, 2020, which are hereby incorporated herein by reference in their entireties.

FIELD OF THE INVENTION

[0002] The present invention relates generally to vehicular vision systems that display video images derived from video image data captured by one or more cameras of the vehicle.

BACKGROUND OF THE INVENTION

[0003] It is known to provide an automotive exterior rearview mirror assembly that includes a foldable mirror assembly, such as a powerfold mirror where the mirror head is pivotable via an actuator between a drive or use position and a folded or park position. It is also known to provide a video display at the exterior rearview mirror assembly, such as described in U.S. Pat. No. 7,777,611, which is hereby incorporated herein by reference in its entirety, or to provide a video display at an interior rearview mirror assembly to display sideward and/or rearward images captured by exterior viewing cameras, such as described in U.S. Pat. No. 5,670,935, which is hereby incorporated herein by reference in its entirety.

SUMMARY OF THE INVENTION

[0004] A pivotable wing or pivotable arm assembly (that supports a wing-mounted camera of a camera monitoring system) includes a powerfold actuator to pivot the pivotable arm or wing and wing-mounted camera of the camera monitoring system between a drive or use position and a folded or park position responsive to a user input.

[0005] The actuator provides a low profile package for powerfold action of the wing-mounted camera with a double detent design for lift and turn action of the camera wing during pivoting of the camera wing between at least a folded and a drive position, either via the motor of the actuator assembly or via manual rotation. The actuator includes an upper detent interface that locks or fixes an output gear relative to the post base structure and a lower detent interface that enables the lift and turn action of the housing of the actuator. The upper detent interface may be provided by the

engagement of an upper detent surface of a middle detent element and a lower detent surface of an upper detent element and is biased into engagement via an internal or integrated biasing element (such as a wave spring) pressing upward on the middle detent element to engage the middle detent element with an upper detent surface of the pivot post base structure provided by the upper detent element. The internal biasing element may be positioned between the output gear and the middle detent element along the post base structure to urge the middle detent element upward along the post base structure to bias the upper detent interface into engagement and to urge the output gear downward along the post base structure to bias the lower detent interface into engagement. At least a portion of the middle detent element and the entirety of the biasing element may be disposed within a cavity of the output gear to provide a low profile detent assembly of the actuator.

[0006] Optionally, the internal biasing element may include a coil spring disposed in the post base structure. In such an example, a pivot tube may include a base portion that extends above the post base structure and a retaining element at a distal end of the pivot tube that retains the biasing member between an inner upper surface of the post base structure and the retaining element. The biasing force of the biasing member pulls the pivot tube downward along the post base structure, pulling the base portion of the pivot tube toward the upper end of the post base structure and biasing an upper detent element downward into engagement with the upper detent interface and biasing the output gear downward into engagement with the lower detent interface.

[0007] Optionally, the camera wing may be pivotable between the folded and drive positions via operation of a powerfold actuator that includes a detent assembly attached at the wing and a cable drive actuator that is operable to pivot the detent assembly to pivot the wing. The cable drive actuator may be disposed remote from the detent assembly and may pull and release tension in a cable to pivot the wing between positions. The detent assembly includes a primary detent that is engaged when the cable drive actuator is operated to pivot the wing between the folded position and the drive position and a secondary detent that is engaged to maintain the wing at the drive position when the wing is manually pivoted to the drive position. The detent assembly may also include a locking component that is engaged when the wing is manually pivoted between the folded position and the drive position and the primary detent is not engaged.

[0008] Optionally, the camera wing may be manually pivotable beyond the drive position to a forward fold position. When the wing is pivoted from the drive position toward the forward fold position, the wing may be at least partially received in a forward cavity at the exterior portion of the vehicle. A forward flap assembly may seal the forward cavity when the wing is between the folded position and the drive position and accommodate the pivotal movement of the wing into the forward cavity when the wing is pivoted toward the forward fold position. The forward flap assembly may bias the wing from the forward fold position toward the drive position.

[0009] These and other objects, advantages, purposes and features of the present invention will become apparent upon review of the following specification in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a plan view of a vehicle having a vision system and wing-mounted camera devices;

[0011] FIG. 2 is an exploded perspective view of an actuator of the wing-mounted camera device;

[0012] FIG. 3 is an exploded perspective view of the output gear, lower housing, and base post structure of the actuator of FIG. 2;

[0013] FIG. 4 is a perspective view of the post base structure of the actuator;

[0014] FIG. 5 is a perspective view of the post base structure disposed at an upper detent element;

[0015] FIGS. 6 and 7 are perspective views of the lower housing of the actuator;

[0016] FIGS. **8** and **9** are perspective view of the output gear of the actuator;
[0017] FIG. **10** is a perspective view of the output gear disposed at the post base structure;
[0018] FIGS. **11** and **12** are perspective views of a middle detent element of the actuator;
[0019] FIGS. **13** and **14** are perspective views of the upper detent element of the actuator;
[0020] FIG. **15** is a sectional view of the actuator;
[0021] FIGS. **16-18** are sectional views of the actuator in different detent states;
[0022] FIGS. **19** and **20** are perspective views of the actuator with the upper housing removed from the lower housing;
[0023] FIG. **21** is a perspective view of a gear train of the actuator;
[0024] FIGS. **22** and **23** are side and top views, respectively, of the gear train of FIG. **21**;
[0025] FIGS. **24** and **25** are exploded views of another actuator;
[0026] FIG. **26** is a perspective view of the gear train of the actuator of FIGS. **24** and **25**;
[0027] FIGS. **27** and **28** are side and top views, respectively, of the gear train of FIG. **26**;
[0028] FIG. **29** is a perspective view of the actuator of FIGS. **24** and **25**;
[0029] FIGS. **30-32** are sectional views of the actuator of FIGS. **24** and **25**;
[0030] FIG. **33** is a perspective view of a camera wing in an extended position at the side of a vehicle;
[0031] FIG. **34** is an exploded view of a camera wing with a detent assembly and a forward fold flap assembly;
[0032] FIGS. **35A-C** are perspective views of the camera wing of FIG. **34** in a flush position, an extended position, and a forward fold position, respectively;
[0033] FIG. **36** is a perspective view of the camera wing of FIG. **34** with a cable pull actuator operable to pivot the camera wing via a cable;
[0034] FIG. **37** is a side view of the cable pull actuator of FIG. **36**;
[0035] FIGS. **38-40** are perspective views of the cable pull actuator of FIG. **36**;
[0036] FIGS. **41A-D** are side views of the cable pull actuator during different stages of its rotation;
[0037] FIG. **42** is a perspective view of the camera wing and detent assembly of FIG. **34**;
[0038] FIG. **43** is a perspective and sectional view of the camera wing and detent assembly;
[0039] FIG. **44** is a sectional view of the detent assembly;
[0040] FIG. **45A** is a sectional view of an upper portion of the detent assembly;
[0041] FIG. **45B** is a sectional view of a lower portion of the detent assembly;
[0042] FIGS. **46-48** are exploded views of portions of the detent assembly;
[0043] FIG. **49** is a perspective view of a locking lug component of the detent assembly;
[0044] FIG. **50** is a perspective view of a primary detent component of the detent assembly;
[0045] FIG. **51** is a perspective view of a secondary detent component of the detent assembly;
[0046] FIG. **52** is a flowchart showing the engagement of the detent assembly based on the position of the camera wing;
[0047] FIG. **53** is a perspective view of a camera wing in a forward fold position;
[0048] FIG. **54** is a perspective view of the camera wing of FIG. **53**, showing a forward fold flap assembly;
[0049] FIG. **55** is a perspective view of the camera wing and forward fold flap assembly, showing the camera wing in a flush position;
[0050] FIG. **56** is a perspective view of the forward fold flap assembly
[0051] FIG. **57A** is a perspective view of the camera wing of FIG. **53**, shown in its retracted position;
[0052] FIG. **57B** is a perspective view of the camera wing of FIG. **53**, shown in its retracted position;
[0053] FIG. **58A** is a perspective view of the camera wing of FIG. **53**, shown in its extended position as extended via actuation of the cable pull actuator;
[0054] FIG. **58B** is a perspective view of the camera wing of FIG. **53**, shown in its extended

position;

[0055] FIG. 59A is a perspective view of the camera wing of FIG. 53, shown in its forward fold position when manually pivoted from the extended state; and

[0056] FIG. 59B is a perspective view of the camera wing of FIG. 53, shown in its forward fold position.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0057] A vehicle vision system and/or driver or driving assist system and/or object detection system and/or alert system operates to capture images exterior of the vehicle and may process the captured image data to display images and to detect objects at or near the vehicle and in the predicted path of the vehicle, such as to assist a driver of the vehicle in maneuvering the vehicle in a rearward direction. The vision system includes an image processor or image processing system that is operable to receive image data from one or more cameras and to provide an output to one or more display devices for displaying video images representative of the captured image data. For example, the vision system may provide a rearview display (such as derived from image data captured by one or more rearward viewing cameras at the vehicle) or a top down or bird's eye or surround view display or the like.

[0058] Referring now to the drawings and the illustrative embodiments depicted therein, a vehicle **10** includes an imaging system or vision system **12** that includes multiple exterior viewing cameras, including wing-mounted pivotable camera devices or modules **11** at each side of the vehicle (FIG. 1), such as at or in place of traditional exterior rearview mirror camera locations. The wing-mounted cameras **14** may be disposed on pivotable arms or camera wings **13** pivotable relative to a mounting base or arm **15** between a use or drive position and a park or folded position where at the drive position, the pivotable arm or camera wing **13** is extended from the side of the vehicle so that the camera **14** may view sideward and rearward of the vehicle to capture image data representative of a view sideward and rearward of the vehicle. At the park position, the camera wing **13** is pivoted in toward (and partially into) the respective side of the vehicle (and may be optionally received within a recess at the respective side of the vehicle). The wing-mounted camera device **11** is pivotable between the drive and park positions via operation of an actuator **20** that may be operated responsive to a user input. The wing-mounted camera device may also be manually pivotable between the drive and park positions. The camera wing and wing-mounted camera may have features similar to those described in International Publication No. WO 2021/263280, which is hereby incorporated herein by reference in its entirety.

[0059] As will be described further below, the actuator is operable to pivot the wing between at least the drive and park positions and includes a biasing element such as a coil spring or wave spring accommodated substantially or entirely within an output gear to provide a self-contained and low-profile, compact double detent actuator that is disposed in thin or low profile wing structures and that enables lift and turn pivoting of the wing. The actuator includes an output gear and when the wing is pivoted between the drive and park positions via the actuator, the housing of the actuator travels around the output gear fixed relative to a base of the actuator. When the wing is manually pivoted between positions, the output gear is fixed relative to the wing and thus rotates about a post portion of the base as the wing pivots. The actuator further includes the internal spring that is accommodated substantially or entirely within a cavity of the output gear of the actuator and that provides a biasing force on the output gear and a middle detent element (that provides a portion of the upper detent interface of the actuator). Additionally, at least a portion of the middle detent element is received or accommodated within the cavity of the output gear.

[0060] The system **12** may also include other cameras, such as surround view cameras (including a rearward viewing or rear backup camera, a forward viewing camera at the front of the vehicle and side surround view cameras at respective sides of the vehicle). The side-mounted camera modules **11** may be part of a camera monitoring system (CMS), which may provide a rearward viewing camera that has a different field of view than the rear backup camera, with the camera **13** (when the

module is extended) capturing image data of the respective scene exterior of the vehicle and in the field of view of the respective camera, with each camera having a lens for focusing images at or onto an imaging array or imaging plane or imager of the camera. The system **12** may display video images derived from image data captured by the cameras at a video display screen **18** in the vehicle for viewing by the driver of the vehicle while the driver is operating the vehicle **10**.

[0061] The vision system **12** includes a control or electronic control unit (ECU) **16** having electronic circuitry and associated software, with the electronic circuitry including a data processor or image processor that is operable to process image data captured by the cameras, whereby the ECU **16** may detect or determine presence of objects or the like and/or the system **12** may provide video images to the display device **18** of the interior rearview mirror assembly **19** of the vehicle for viewing by the driver of the vehicle and/or to a display device at the center console or stack of the vehicle (and optionally to CMS displays at or near the driver and passenger side A-pillars of the vehicle, such as described in U.S. Publication Nos. US-2018-0134217 and/or US-2014-0285666, which are hereby incorporated herein by reference in their entireties). The data transfer or signal communication from the cameras to the ECU may comprise any suitable data or communication link, such as a vehicle network bus or CAN (Controller Area Network) bus or LIN (Local Interconnect Network) bus or **12C** bus or the like of the equipped vehicle.

[0062] The rearward viewing cameras of the CMS cameras may function to provide rearward video images for a dual-mode interior rearview video mirror that can switch from a traditional reflection mode to a panoramic live-video display mode. The system and cameras and display device may utilize aspects of the systems described in U.S. Pat. Nos. 11,242,008; 10,442,360; 10,421,404; 10,166,924; 10,046,706; 9,676,336 and/or 9,487,142, and/or U.S. Publication Nos. US-2021-0162926; US-2021-0155167; US-2021-0094473; US-2020-0377022; US-2019-0258131; US-2019-0146297; US-2019-0118717; US-2019-0047475 and/or US-2017-0355312, and/or International PCT Application No. PCT/US2022/070062, filed Jan. 6, 2022 and published as International Publication No. WO 2022/150826, which are all hereby incorporated herein by reference in their entireties.

[0063] The wing-mounted pivotable camera device **11** includes a wing **13** pivotally mounted at a base portion **15** (such as a mounting base or mounting arm), the pivotable wing accommodating a camera **14** operable to capture image data. The wing or camera wing **13** is pivotable between a use or drive position (where the wing is extended from the side of the vehicle so that the camera is extended and/or spaced from the side of the vehicle and has a field of view at least sideward and rearward of the vehicle along the respective side of the vehicle at which the device is mounted) and a park or folded position (where the wing is folded in toward the respective side of the vehicle and the camera is unable to view along the side of the vehicle). As shown in FIG. **1**, one wing-mounted pivotable camera device **11** is disposed at each side of the vehicle **10** to provide sideward and rearward views along each side of the equipped vehicle. The base portion or mounting arm **15** is disposed at the side of the vehicle **10** with the camera wing **13** pivotally attached thereat and the device **11** is a powerfold assembly that includes the actuator **20** operable to pivot the camera wing **13** relative to the mounting arm or base **15**. Additionally, the wing-mounted pivotable camera device **11** may comprise a breakaway device (where the camera wing **13** may be manually pivoted relative to the mounting arm or base **15**). Although the actuator described herein is described in reference to a wing-mounted camera, it should be understood that the actuator may be compatible with exterior rearview mirror assemblies as well as other rotatable mounting structures and exterior mounted components.

[0064] The actuator **20** operates, responsive to a user input, to pivot the wing **13** between a plurality of detent positions, including the use or drive position (where the wing **13** is extended laterally from the side of the vehicle **10** and the camera **14** views generally rearward along the side of the vehicle) and the folded or park position (where the wing **13** is pivoted to be generally along the side of the vehicle **10** and optionally recessed within the side of the vehicle at which the wing is

mounted). The wing **13** is also pivotable manually to either the use or folded position. Optionally, the wing **13** may be pivotable beyond the drive position to a fully forward or open position. A seal may be disposed along the interface between the wing **13** and the mounting arm or base **15**. The actuator may utilize aspects of the actuators described in U.S. Pat. Nos. 7,887,202 and/or 9,487,142, and/or U.S. Publication Nos. US-2021-0261053 and/or US-2020-0223364, which are all hereby incorporated herein by reference in their entireties.

[0065] The powerfold actuator **20** has a self-contained compact, pivot, double detent design and is capable of having an all-plastic gear train (discussed further below) with spur gear reduction for added strength and reduced wear, which may provide a flexible gear train that allows for more freedom in packaging design and gear ratio tuning and reduced thrust on motor shaft to improve longevity and reduce motor “click.” The double detent design enables lifting or raising of the wing **13** (which is attached to the housing of the actuator) when the mounting structure is rotated to reduce stress on the seal disposed between the wing **13** and the mounting arm or base **15** and allows the actuator **20** to return to a nominal state after the mounting structure is manually rotated.

[0066] With reference to FIGS. 2-23, the actuator **20** comprises a housing **22** that accommodates an electrically operable motor **24**, a gear train **26**, and a detent assembly disposed at a base post structure **28** that has a pivot tube or post **30** and base **32** integrated together with the pivot post providing or defining a pivot axis for the wing **13**. The post **30** extends from the base **32** and an output gear **34** is disposed along the post **30** at the bottom of the post (i.e., at or near the base **32**), with the post **30** extending through a central axis of the output gear **24**. An actuator housing or lower housing **23** is rotatably disposed at the post **30** and houses the motor **24** and is non-rotatably disposed or attached at the camera wing **13**. The post **30** extends from the base **32** and through the output gear **34** and lower housing **23** and through a middle detent element **36** and at least a portion of an upper detent element **38**. An upper housing portion **21** attaches to the lower housing **23** and encloses the base post, motor, detent elements, output gear and gear train. The base **28** is fixed at the mounting arm **15** (such as via bolts) and the housing is fixed relative to the wing **13** so that when the actuator is operated, the housing (and therefore the wing) pivots relative to the base or mounting arm **15**. The actuator **20** also enables manual pivoting of the wing **13** relative to the mounting arm **15** via the detent assembly (collectively the output gear, upper detent element or part, middle detent element or part, and a biasing element or wave spring **52**) and its engagement with the housing so that, if the wing **13** is manually rotated, the actuator **20** may return or reset to a nominal position prior to subsequent motor driven operation of the actuator.

[0067] The upper detent **38** is non-rotatably keyed with the base structure **28** such that the upper detent **38** is non-rotatable relative to the post (FIG. 5). Thus, both the base structure **28** and the upper detent **38** are fixed and cannot move relative to one another and the base, pivot post and upper detent can be riveted or otherwise fastened together. The middle detent **36** is keyed with the output gear **34** such that the middle detent **36** and output gear **34** rotate together and at least a portion of the middle detent **36** is accommodated or received within a cavity of the output gear **34**. The middle detent **36** and output gear **34** may independently travel vertically or longitudinally along the pivot post **30** and are biased or urged apart by the wave spring **52** disposed between the middle detent **36** and output gear **34** and accommodated or disposed entirely within the cavity of the output gear **34**.

[0068] The actuator **20** has two detent interfaces that become active (i.e., engage to affect operation of the actuator **20**) at different times. The upper detent interface is between an upper surface of the middle detent element **36** and a lower surface of the upper detent element **38**. The lower detent interface is between a lower surface of the lower housing **23** and the top or upper surface of the base **32**. A third interface may be referred to as the ramp interface, which is between a lower surface or ramp structure of the output gear **34** and the top or upper surface of the base portion **32** of the base post structure **28**. As described below, the upper detent interface, the lower detent interface, and the ramp interface enable the detent elements, output gear, and lower housing to

engage and disengage with one another so that the housing **22** and wing **13** may lift and pivot relative to the mounting arm **15** between the drive and fold positions (and optionally an open position) either as driven by the motor **24** of the actuator **20** or as manually pivoted by a user. [0069] The base post structure **28** with base portion **32** and pivot post **30** is fixed to the mounting arm **15** with the pivot post **30** providing the pivot axis for the wing **13** and actuator **20**. The base portion **32** includes a bottom surface (that may engage a surface of the mounting arm or base when mounted thereat) and the top surface opposite the bottom surface that engages the bottom surface of the lower housing **23**. As shown in FIGS. **4** and **5**, the base post structure **28** is keyed with the upper detent part **38** via a series of protrusions **40** circumscribing the pivot post **30** to fix the upper detent part **38** relative to the base **28** and prevent the upper detent **38** from rotating about the pivot post **30**. Circumscribing the outer radius of the post protrusions **40** is a ring of lift ramps **42** configured to engage corresponding ramp structure of the output gear **34** (collectively forming the ramp interface). The lift ramps **42** guide vertical (upward and downward) movement of the output gear **34** along the pivot post **30** as the output gear rotates about the pivot post **30**. Disposed radially outward from the lift ramps **42** is a grouping of detents **44** (referred to as base detents) at the upper end or surface of the base portion **32**. Base detents **44** are configured to engage corresponding detents **25** at the lower end or surface of the lower housing **23** so that, when the base detents **44** are aligned with the detents **25** of the lower housing **23** (i.e., the detents are nested with one another, such as in FIG. **16**), the housing **22** (and thus camera wing) is in a lowered state, and when the detents are not aligned (i.e., rotated so as to engage one another, such as in FIG. **17**), the housing **22** (and thus camera wing) is in a raised state. The detents are aligned or nested when a raised portion of one detent engages a lowered portion of another detent. Thus, the housing **22** is in a lowered state when the camera wing is in a final (drive or park) position and the housing is in a raised state when the wing is being pivoted between the final positions.

[0070] The bottom surface of the lower housing **23** (such as seen in FIG. **6**) includes the detents **25** (referred to as housing detents) that are configured to engage the base detents **44** of the base portion, thus forming the primary or lower detent interface. When the lower housing **23** is rotated (as described further below), the raised portion of the housing detents **25** lift out of the corresponding lowered portion of the base detents **44** (as guided by the slanted or sloped surfaces of the detents) and move along the raised portion of the base detents **44** to maintain the housing at the lifted position as the housing rotates to its next position (such as from the drive position to the park position). The lower housing **23** also has an upper or inner surface **27** configured to engage a bottom surface of the output gear **34**.

[0071] As shown in FIG. **7**, the lower housing **23** includes the cavity in which the motors, gear train, output gear and detent elements are accommodated so that the entire apparatus may be a reduced or lower profile assembly. The upper or inner surface **27** of the lower housing cavity engages a lower surface of the output gear **34** and includes protrusions or stops **27a** to engage corresponding structure on the bottom surface of the output gear **34**. The stop protrusions **27a** provide a resistant element so that as the camera wing **13** pivots between the drive and park positions, the stops **27a** may engage the corresponding structure on the bottom surface of the output gear **34** to preclude further movement of the housing relative to the output gear **34** and stall the motor. That is, the position of the stops **27a** correspond to the position of the wing **13** at the drive and park positions so that the motor is stalled and the housing and wing may cease pivoting when the motor is electrically operated to pivot the camera wing to the drive or park position.

[0072] The output gear **34** is disposed at the pivot post and is selectively pivotable (as described below) about the pivot post **30** based on engagement of the upper detent interface. The pivot post **30** extends from the base portion **32** through a central through hole of the output gear **34**. As shown in FIG. **8**, protruding from the lower end or surface of the output gear and circumscribing the central through hole is a ramp structure **46** configured to engage the lift ramps **42** of the base structure (when the output gear is disposed at the pivot post) so that as the output gear **34** rotates

about the pivot post **30**, the ramp structure **46** engages the lift ramps **42**, causing the output gear **34** to lift relative to the base portion **32** along the pivot post **30**. When the output gear **34** is disposed at the pivot post **30**, the ramp structure **46** protrudes through an opening in the lower housing **23** (through which the pivot post also extends) to engage the lift ramps **42** of the base structure. Optionally, the lift ramps **42** may protrude through the opening in the lower housing **23** to engage the ramp structure **46** of the output gear **34** within the housing. When a peak of the ramp structure **46** is engaged with a peak of the lift ramp **42**, the output gear **34** is in a raised position along the pivot post **30** relative to the base structure **28** (FIG. **10**). When the peak of the ramp structure is in a valley of the lift ramp, the output gear is in a lowered position along the pivot post relative to the base structure.

[0073] As shown in FIG. **9**, a plurality of protrusions **48** (referred to as output gear stops) are also disposed at the bottom surface of the output gear **34** and are configured to engage the stops **27a** at the upper or inner surface **27** of the lower housing **23** to guide rotation of the lower housing **23** relative to the output gear **34**. When the lower housing **23** rotates about the output gear **34**, the stops **27a** of the lower housing **23** may engage the output gear stops **48** to provide the hard stop and stall the motor **24**. The stops **27a** are configured so that the output gear **34** is in a cinch position with the lower housing **23** when the camera wing **13** is in the drive position. Furthermore, the output gear stops **48** have a sloped surface so that during certain rotations, the output gear may rotate past engagement with the lower housing stops **27a**.

[0074] The output gear **34** has a cavity that receives the wave spring **52** and at least a portion of the middle detent **36**. The output gear **34** is keyed together with the middle detent **36** via a series of protrusions **50** (FIG. **9**) circumscribing the through hole of the output gear and corresponding protrusions **56** from a bottom surface of a flange of the middle detent **36** so that the middle detent **36** is rotationally fixed relative to the output gear **34**. The middle detent **36** and output gear **34** may travel independently in a longitudinal direction along the pivot post, but rotate together due to the keyed protrusions. The wave spring **52** engages the inner surface of the output gear cavity and the bottom surface of the flange of the middle detent **36** and is sandwiched therebetween to impart a separating biasing force between the two components. Thus, the wave spring **52** biases or urges the output gear **34** and the middle detent **36** in opposite, separating directions so that the output gear **34** is biased toward the lower housing **23** (for example, so that the ramp structure **46** of the output gear **34** engages the lift ramps **42** of the base structure) and the middle detent **36** is biased into engagement with the upper detent **38**. Thus, the wave spring **52** provides an internal or integrated spring within the detent assembly to further reduce the profile of the actuator **20**.

[0075] The middle detent **36** (FIGS. **11** and **12**) comprises a flange **54** with a top and bottom surface and protrusions **56** extending from the bottom surface of the flange **54** to engage the corresponding protrusions **50** of the output gear. The top surface of the flange **54** includes a series of detents **58** (referred to as middle detents) configured to engage corresponding detents on the bottom surface of a flange of the upper detent element **38**, thus forming the upper detent interface. Because at least a portion of the middle detent element **36** is disposed within the cavity of the output gear **34**, the upper detent interface may also engage within or in close proximity to the cavity of the output gear **34**. This further reduces the profile of the actuator **20**.

[0076] The upper detent **38** (FIGS. **13** and **14**) comprises a flange **60** with a top and bottom surface, a post structure **62** extending from the bottom surface of the upper detent flange **60** and protrusions **64** extending from the post structure **62** engaging corresponding structure at the base structure **28** to rotationally fix the upper detent **38** to the base structure **28**. The post structure **62** of the upper detent **38** extends along the pivot post **30**. The bottom surface of the flange **60** of the upper detent **38** includes detents **66** (referred to as upper detents) corresponding to the middle detents of the middle detent element **36**. When the upper detent **38** engages the middle detent **36** to form the upper detent interface, an upper end or surface of the upper detent flange **60** effectively caps or retains the wave spring **52** and the middle detent **36** within the cavity of the output gear **34**. For

example, see FIG. 2 where the output gear assembly (with upper detent encasing the middle detent and wave spring within the cavity of the output gear) provides a singular, low profile and compact construction to be disposed in the housing. Thus, the middle detent element, upper detent element, wave spring, and output gear contribute to the low profile nature of the actuator.

[0077] The upper housing **21** attaches to the lower housing **23** to house the pivot or detent assembly (collectively the output gear, wave spring, middle detent, and upper detent) and motor or drive assembly (collectively the motor and gear train) within a cavity defined by the upper and lower housing. The upper housing **21** is fixed to the lower housing **23** (such as via bolts, adhesive, or the like) and rotates with the lower housing **23**. As shown in FIG. 15, the actuator provides a compact assembly to be disposed in a narrow or thin wing-mounted camera wing. The lower housing **23** provides a cavity that accommodates the detent assembly and the output gear **34** of the detent assembly provides a cavity that accommodates at least a portion of the middle detent, spring, and upper detent. The output gear **34** engages the base structure at a ramp interface, the lower housing **23** engages the base structure at a lower detent interface and the middle detent **36** engages the upper detent **38** at an upper detent interface. Collectively, the housing rotates about the pivot post **30**, as discussed below, either via manual rotation or under power of the drive assembly through the detent assembly (which may be referred to as a power fold rotation or event). With the housing fixed relative to the camera wing **13**, the camera wing **13** pivots about the pivot post **30** according to movement of the actuator housing.

[0078] When the camera wing **13** is rotated between the drive and fold positions via operation of the motor **24**, the output gear **34** is fixed relative to the pivot post **30**. For example (as shown in FIG. 16), the actuator may be positioned in a drive position, where the housing (and therefore camera wing) is lowered along the pivot post relative to the base portion with the lower detent interface engaged. Thus, the lower or recessed portions of the base detents **44** are aligned or keyed with the raised portions of the lower housing detents **25** and vice versa. The wave spring **52** imparts a biasing force to push the middle detent **36** into engagement with the upper detent **38** (engaging the upper detent interface) and the bottom surface of the output gear **34** toward the upper or inner surface **27** of the lower housing **23**. Engagement of the upper detent interface is the alignment of the raised portion of upper detents **66** with the lowered or recessed portions of the middle detents **58** and vice versa. Thus, due to the engagement of the upper detent interface, when the motor **24** provides rotational force via the gear train **26** to the output gear **34** (such as to power fold the camera wing to the park position), the output gear **34** is fixed relative to the pivot post **30** via the middle detent **38** being rotationally fixed to the output gear and engaged with the upper detent **38** and the gear train **26** travels around the fixed output gear **34** to rotate the housing about the pivot post **30**. In other words, the operation of the motor **24** turns the gears of the gear train **26** engaging the output gear and, because the output gear **34** is rotationally fixed, the rotational force is imparted on the housing to pivot the housing and camera wing **13**. Thus the load path is through the output gear and lower housing.

[0079] When in the drive position, before the actuator **20** is actuated and without any manual pivoting (FIG. 16), the output gear **34** is held up or in a raised position along the pivot post **30** relative to the base structure via the lift ramps and ramp structure and the stops **27a** of the lower housing **23** are engaged with the output gear stops **48** to cinch the output gear **34** relative to the housing (the nominal or starting position). As the housing is rotated toward the folded or park position via operation of the motor **24**, the housing **23** lifts in relation to the base **32** while turning due to the disengagement of the lower detent interface, and such lifting lifts the camera wing **13** and increases the gap between the camera wing **13** and the mounting base **15** and thus reduces or limits or avoids pressure at the cut line seal (the seal that is disposed between and at the interface of the wing and the mounting arm and that follows the contour and fills the gap between the camera wing and the mounting arm) during pivotal movement of the camera wing **13**. The lower housing **23**, as it lifts up, is provided clearance to rotate because the output gear **34** is in the raised position

via the ramp interface and rotationally fixed relative to the pivot post **30** via the upper detent interface (and so does not lower while the housing rotates about the pivot post). The biasing force of the wave spring **52** maintains the engagement of the upper detent interface during rotation of the housing about the output gear **34**. As can be seen with reference to FIG. **17**, the lower housing **23** and base detents thus disengage and the lower housing **23** lifts up and rotates at a lifted position due to the raised surface of the lower housing detents **25** traveling along the raised surface of the base detents **42**.

[0080] When in the park or powerfold position with the output gear **34** in the raised position via the lift ramps (such as after being rotated from the drive position to the powerfold position via the motor), the camera wing **13** is rotatable to the drive position either via activation of the motor **24** or via manual rotation. If rotated from the powerfold position to the drive position via activation of the motor **24**, the output gear **34** remains fixed relative to the base portion **28** (via engagement of the upper detent interface) and the housing travels around the output gear **34** until the lower detent interface engages and the housing is in the cinch position relative to the output gear **34** via engagement of the stops **27a**.

[0081] Thus the motor **24** operates to rotate the housing from the drive position to the park position and from the park position to the drive position. The motor **24** operates until it is stalled or stops, such as when a stop **27a** of the lower housing **23** engages the corresponding protrusion on the underneath surface of the fixed output gear **34**. After the motor **24** is stalled when the camera wing **13** reaches the drive or powerfold position, the next time the motor **24** is activated, it will automatically rotate in the opposite direction to return the camera wing **13** to the previous position. For example, the motor **24** may be operated to rotate from the drive position to the powerfold or park position and as the camera wing **13** approaches the park position, the lower detent surface moves to reengage and the motor **24** is stalled. The next time the motor **24** is actuated, if the housing has not been manually rotated between motor activations, the motor **24** will rotate in the opposite direction to rotate the camera wing **13** from the park to the drive position, at which point the motor **24** will be stalled when the stop **27a** of the lower housing **23** engages the output gear stop **48**. If the camera wing **13** is manually rotated and then the motor **24** is activated, the motor **24** may rotate the output gear **34** back to a nominal starting position where the output gear **34** is raised relative to the base portion **28** and the upper detent interface is engaged and the actuator **20** may then be operational to rotate from its current position to the next position. That is, after manual pivoting, the output gear **34** is not rotationally fixed relative to the pivot post **30** and operation of the motor **24** rotates the output gear about the pivot post **30**.

[0082] For example, if the camera wing **13** is in the drive position, with the output gear **34** at the raised position, and the actuator **20** is manually pivoted from the drive position to the park position, the rotational force is transferred (via locked gears of the gear train **26** connected to the motor **24**) from the lower housing **23** to the output gear **34**. Thus, the output gear **34** begins to rotate and the upper detent interface disengages as the output gear ramp structure **46** travels down the lift ramps **42** to lower the output gear **34** along the pivot post **30** to provide clearance for the middle detent **36** to lower relative to the upper detent **38** against the biasing force of the wave spring **52**. The housing (and thus camera wing) resist further rotation from the drive position beyond the park position when the lower detent interface reengages at the park position.

[0083] If rotated from the park position to the drive position manually (with the output gear **34** in a raised position), the housing and output gear **34** are fixed relative to one another (via the locked gear train **26**) and rotate together about the pivot post **30**. The output gear **34** lowers along the pivot post **30** as the ramp structure **46** travels down the lift ramp **42** while the output gear **34** rotates, thus providing clearance for the upper detent interface to disengage. The housing is precluded from rotating beyond the drive position when the lower detent interface reengages and the output gear **34** is in the lowered position relative to the base **28**.

[0084] After being manually rotated from the park to drive position, however, the output gear **34** is

not in the cinch position with the lower housing 23. As shown in FIG. 18, the stop 27a of the lower housing is not aligned with the output gear stop 48 and the output gear 34 has been moved down the lift ramp 42 so that the load path is again through the output gear 34 and lower housing 23. In this position, the upper detent interface is disengaged. When the motor 24 is next operated, the motor 24 rotates the output gear 34 (and not the housing) about the pivot post 30 because the upper detent interface is disengaged which enables movement of the output gear 34 via the motor 24. The output gear 34 climbs the lift ramp 42 to re-align the stop 48 of the output gear 34 with the stop 27a of the lower housing 23 and to reengage the upper detent interface so that the output gear 34 returns to a cinch position with the lower housing 23 and the lower and upper detent interfaces are engaged. Thus, the output gear 34 and middle detent 36 return to a nominal position and the actuator 20 may be operated to rotate from the drive position to the fold position.

[0085] The camera wing 13 may be rotated between the drive and park positions via the motor 24, which is operable to impart rotational forces on the output gear 34 via a gear train 26 (FIGS. 19-23). The gear train 26 is optimized for longer, narrower wing or mounting arm designs with different gear ratio combinations and alternative configurations possible. When the motor 24 is not powered, the gears 26 are effectively locked so that rotational force placed on the housing is imparted on the output gear 34 rather than winding or reversing the motor 24. In the illustrated embodiment, the motor 24 is actuated to rotatably drive a motor pinion 24a that rotatably drives a spur element 68a of a low pressure worm gear 68, which engages a first combo gear 70 which engages a second combo gear 72 which engages the output gear 34 and rotates the housing relative to the output gear until a hard stop is engaged, whereby the motor will stall or stop, with the camera wing retained at the extended or drive position via the detents. The actuator 20 is configured such that a hard stop is provided at the appropriate locations when the camera wing 13 is pivoted via driving of the motor to retain the camera wing 13 at the drive and/or park positions.

[0086] The spur element 68a of the low pressure worm gear 68 may have a gear ratio relative to the motor pinion 24a of 50 to 10 or 5 to 1. The first step 70a of the first combo gear 70 may have a gear ratio relative to the low pressure worm gear 68 of 30 to 1. The first step 72a of the second combo gear 72 may have a gear ratio relative to the second step 70b of the first combo gear 70 of 30 to 10 or 3 to 1. The output gear 34 may have a gear ratio relative to the second step 72b of the second combo gear 72 of 62 to 15 or 4.13 to 1. Thus, the gear train 26 may have a total gear ratio of 1858.5 to 1 from the motor pinion 24a to the output gear 34. The spur gear train may provide a reduced friction, rolling interface with a large contact area and reduced pressure on the gear elements. Thus, there is a reduced material hardness requirement which allows for use of all plastic gears. Other suitable hard and durable materials (such as steel or aluminum) may be used. The spur gear train allows for greater gearbox packaging flexibility and simplified gear tooling and molding. The tight packaging allows for improved constraint of gears within and by the upper and lower housings. Additionally, there is no thrust load on the output gear from the driving gear and reduced thrust on the motor shaft, improving longevity of the motor and reducing motor "click."

[0087] Thus, the actuator provides a low profile, double detent design that enables lift and turn rotation of a camera wing mounted at a vehicle. The actuator includes an integrated or internal spring, a primary or lower detent interface that maintains the camera wing at a lowered drive position and a secondary or upper detent interface that locks the output gear relative to the base portion so that the housing may travel around the output gear when the motor is activated. The output gear may travel vertically along the pivot post via engagement at the ramp interface to enable manual pivoting of the camera wing. The internal spring biases the upper detent interface into engagement and the output gear downward toward the base portion to enable easier disengagement of the upper detent interface when manually pivoting the camera wing. The internal spring may be accommodated entirely within a cavity of the output gear and at least a portion of the middle detent element (forming a portion of the upper detent interface) may also be accommodated within the cavity of the output gear.

[0088] When the motor is operated to rotate the camera wing, the upper detent interface is engaged to fix the output gear relative to the base post structure and the motor and housing travels around the fixed output gear. As the housing travels around the fixed output gear, the lower detent interface disengages, causing the housing (and therefore camera wing) to lift relative to the base post structure. When the camera wing is manually pivoted, the output gear rotates with the housing relative to the base post structure. The ramp structure of the output gear interfaces with the lift ramp of the base structure and the output gear is guided vertically along the pivot post via the lift ramps when the output gear rotates about the pivot post, thus disengaging the upper detent interface. The cavity of the output gear accommodates at least a portion of the middle detent element (which provides a surface of the upper detent interface) and may accommodate the entirety of the spring. The internal spring biases the upper detent interface into engagement and biases the output gear down the ramp structure to assist in downward travel of the output gear during manual rotation.

[0089] When the motor is activated to rotate the camera wing about the pivot post, the motor travels until it is stalled, such as via engagement of structure of the lower housing engaging the portion of the output gear to provide the hard stop when the actuator is at the drive or folded positions. After being stalled and upon the next activation of the motor, the motor rotates in the opposite direction to return the camera wing to its previous position. If the camera wing is manually pivoted between activations of the motor, the next activation of the motor rotates the output gear back to a nominal position and the next activation of the motor after that will rotate the actuator to its next position. The actuator uses a gear train to connect the motor and output gear to increase durability of the individual gear elements and reduce strain on the motor. When the motor is not powered to rotate the camera wing, the gear train is locked so that a rotational force manually rotating the camera wing may be directed from the housing to the output gear.

[0090] Referring now to FIGS. 24-32, another actuator **120** with a detent assembly and an internal or integrated spring may include a resilient element and an internal coil spring **152**. Similar to the actuator **20** described above, the actuator **120** may be equipped at the vehicle for pivoting the low profile camera wing **13**, low profile exterior mirror, or the like. The actuator **120** provides a lightweight plastic actuator design. As shown in FIG. 32, the base post structure **128** is received in the lower housing **123** and includes a pivot post **130** and a base portion **132** that provides a lower outer surface of the lower housing. The base portion **132** includes a three fastener mounting method for improved stability when mounting at a vehicle. The base post structure **128** receives a pivot tube **129** and the coil spring **152** is disposed entirely within the pivot post **130** and receives a post portion **129a** of the pivot tube **129** therein. A PALNUT™ or retaining element **131** is disposed at the bottom of the post portion of the pivot tube to compress and retain the coil spring entirely within the cavity of the pivot post **130**.

[0091] The pivot post **130** is received in the output gear **134** (with the cam surfaces recessed into the gear) and the upper detent **138** with the coil spring **152** disposed at an interior portion of the pivot post **130**. Thus, the internal coil spring **152** is disposed within the pivot post **130** and at least a portion of the output gear **134** and at least a portion of the upper detent **138** circumscribe the spring. The internal coil spring **152** enables a compact actuator design where the spring is retained within a component of the actuator **120**. A resilient element **133** is disposed around a lower portion of the output gear **134** to provide a hard stop when the camera wing is rotated between the drive and fold positions. The biasing force of the coil spring presses against the internal surface of the pivot post **130** and the retaining element **131** to pull a base portion **129b** of the pivot tube **129** downward toward the top of the pivot post **130**, thus biasing the output gear **134** and upper detent **138** into engagement at an upper detent interface.

[0092] That is, the pivot tube **129** includes a post portion **129a** that extends through a cavity of the pivot post **130** and includes the base portion **129b** exterior and above the pivot post **130**. The spring **152** is disposed within the cavity of the pivot post **130** and is disposed between an inner upper

surface of the pivot post **130** and the retaining element **131** at the distal end of the post portion **129a**. The spring **152** pushes the retaining element **131** away from the fixed surface of the pivot post **130** and thus biases the pivot tube **129** down along the longitudinal axis of the pivot post **130**. This in turn biases the output gear **134** and upper detent **138** downward along the pivot post **130**. [0093] As shown in FIGS. **26-28**, the output gear is engaged by a motor **124** via a main gear **126**. The spur gear portion of the main gear **126** may have a gear ratio relative to the motor worm or pinion **126a** of 34 to 1. The output gear **134** may have a gear ratio relative to the main gear of 40 to 1. Thus the output gear **134** may have a total gear ratio relative to the motor pinion **124a** of 1360 to 1. The lower housing **123** attaches to the upper housing **121** (FIG. **29-32**) and is fixed to the camera wing such that lift and rotation of the housing corresponds to lift and rotation of the camera wing. [0094] Thus, the actuator provides a reduced profile actuator suitable for a camera wing application, with the spring and pivot tube being internal of and below the output gear. The spring biases or urges the pivot tube and the upper detent and the output gear downward toward the base portion. By providing the tube and spring within and below the output gear, the actuator provides a low profile actuator that is suitable for low profile camera wing applications and also suitable for pivoting an exterior mirror head relative to mounting structure.

[0095] Referring now to FIGS. **33-52**, another actuator **220** suitable for a pivotable camera wing application and having a detent assembly with an internal or integrated spring may include a cable pull actuator **224**, a locking lug component **274**, a secondary detent **236** at the top of the camera wing, and a forward fold flap assembly **280**. Instead of being driven directly by a cam or motor via a gear train, the actuator or detent assembly **220** is activated from a remote actuator, such as via a push/pull Bowden cable **226** attached to the cable pull actuator **224**. The cable pull actuator **224** rotates the camera wing **213** by pulling on a cable pull arm **234** connected to the detent assembly. The locking lug component **274** selectively fixes or retains the cable pull arm **234** relative to a fixed bracket **228** at the vehicle to connect rotation of the camera wing with movement of the cable pull actuator **224** during electrical pivoting of the camera wing **213** and disconnect rotation of the camera wing from affecting movement of the cable pull actuator during manual rotation of the camera wing. The secondary detent **236** engages the camera wing **213** when the camera wing is manually pivoted to the extended position to retain the camera wing at the extended position. The forward fold flap assembly **280** allows the camera wing **213** to rotate beyond the extended position and provides a biasing force to return the camera wing to the extended position after being rotated beyond the extended position.

[0096] As shown in FIGS. **35A-C**, the camera wing **213** is movable via the actuator **220** between at least a flush or park position (FIG. **35A**), an extended or drive position (FIG. **35B**) and a forward fold position (FIG. **35C**). When in the extended or drive position **213b**, the camera wing **213** is extended outward from the body of the vehicle at the side of the vehicle so that the camera has a field of view sideward and rearward of the vehicle at the respective side of the vehicle. When in the flush or park position **213a**, the camera wing **213** is pivoted in toward the side of the vehicle so that the camera wing **213** is received in a recess at the side of the vehicle and an exterior surface of the camera wing **213** is generally flush or coplanar with an outer surface of the vehicle. When the camera wing **213** is in the respective park or drive positions, the actuator **220** is configured to lock or hold the camera wing **213** in place. As shown in FIG. **35C**, the camera wing **213** is manually pivotable to the forward fold position **213c**, where the camera wing **213** is rotated beyond the extended position, such as by an impact with an object, and is spring loaded to return to the extended position after being manually rotated beyond the extended position. The forward fold flap assembly **280** enables the camera wing **213** to rotate beyond the extended position and return to the extended position.

[0097] The camera wing **213** is movable between the flush position **213a** and the extended position **213b** via the cable pull actuator **224**. As shown in FIG. **36**, the cable pull actuator **224** is disposed remote from camera wing and detent assembly and provides rotational force at the detent assembly

via a cable **226** attached at a cable pull arm element **234**. As will be discussed below, the cable pull actuator **224** may only be electrically operated to pivot the camera wing **213** from the flush to the extended position. The camera wing **213** may be returned to the flush position from the extended position via a return torsion spring **268**. For example, the cable **226** attached to the cable pull actuator **224** may comprise a metal cable (e.g., steel or other suitable flexible material with sufficient tensile strength) that is only capable of pulling, not pushing, as pushing the cable may result in bending, damaging or causing the cable to pull away from the cable pull arm element **234** at the actuator.

[0098] When operated to rotate the camera wing **213** from the flush position to the extended position, the cable pull actuator **224** has a high initial pull force, such as to break through ice that has accumulated at the outer surface of the camera wing **213**. The cable pull actuator **224** has a variable torque ratio over the course of its travel with a high ratio at the start for a high pull force and lower ratio near the end of its rotation to increase the pull distance of the cable **226**. The pull starts in a spiral region similar to a screw and transitions to a fully rotational movement.

[0099] As shown in FIGS. **37-40**, the cable pull actuator **224** includes a cam or motor operable to rotate a pin **224a** connected to the end of the cable **226**. The pin **224a** is connected to an end of the cable **226** so that as the pin **224a** rotates at the cable pull actuator **224**, the cable **226** wraps or unwraps around a pulley portion. The rotation of the pin **224a** is guided by an arcuate slot of the cable pull actuator **224** and thus the torque provided by the actuator is variable based on the guidance of the slot at a given point in the rotation.

[0100] Positions **1-4** of FIGS. **41A-D** depict the actuator **224** in various stages of its rotation when operated to pivot the camera wing **213** from the flush position to the extended position. When the actuator **224** is activated, the cam pulley rotates counterclockwise. At position **1** (FIG. **41A**), the pin **224a** starts in the spiral region of the cam and slides along the slot in the housing of the cable pull actuator **224** to arrive at position **2** (FIG. **41B**). Roughly half of the total cam rotation happens in this region, yet the pin **224a** moves a very small distance (high ratio, high force, low distance travelled) corresponding with a high initial opening force of the camera wing **213**. Once the pin **224a** hits the end of the spiral region of the cam at position **3** (FIG. **41C**), the cable **226** starts to wind around the pulley portion of the cam. Roughly 80 percent of the total cable pull distance occurs here, where the cam only uses 50 percent of its rotation (low ratio, low force, high distance travelled). At position **4** (FIG. **41D**), the camera wing **213** is at the extended position and the pin **224a** has completed its travel about the pulley portion of the cam.

[0101] Tension in the cable **226** created when the actuator **224** pulls the camera wing **213** to the extended position helps to hold (along with the detent assembly) the camera wing **213** in the extended position. When the camera wing **213** is electrically rotated back to the folded position (i.e., when the actuator rotates in the opposite direction to allow slack in the cable), the actuator **224** begins releasing tension in the cable **226** (i.e., rotates the opposite direction as when extending the wing) and the torsion return spring **268** helps to unwrap the cable **226** from the pulley so that the camera wing **213** may pivot to the folded position. The torsion return spring **268** is used to return the wing to the flush position when the cable pull actuator **224** releases tension on the cable **226**. In other words, the torsion return spring **268** works to rotate the camera wing **213** in a direction opposing the cable pull actuator (i.e., the torsion return spring pushes at the pivot of the camera wing and the cable pull actuator pulls at the pivot of the camera wing via the cable). The torsion return spring **268** may directly engage the cable pull arm **234**.

[0102] When the camera wing **213** is manually rotated between the flush and extended position, the cable pull actuator **224** does not rotate and thus the cable pull arm **234** at the detent assembly remains stationary during manual rotation of the camera wing **213**. If the camera wing **213** is manually rotated to a position without being manually rotated back to its original position, the next activation of the cable pull actuator **224** will resync the cable pull arm **234** according to the position of the camera wing **213** and the following operation of the cable pull actuator **224** will pivot the

camera wing **213**. For example, if the camera wing **213** is manually rotated from the flush to the extended position, the cable pull arm **234** (and cable pull actuator **224**) will remain stationary. Upon the next activation of the cable pull actuator **224**, the cable **226** will pull the cable pull arm **234** into alignment with the detent assembly and the cable pull arm **234** will resync with the camera wing **234** in the extended position and the following activation will release tension in the cable **226** to electrically rotate the camera wing **213** back to the flush position.

[0103] As shown in FIGS. **42-51**, the rotational relationship between the cable pull arm **234** (and thus cable pull actuator) and the camera wing **213** is enabled via the detent assembly of the actuator **220**, which includes a primary detent component **238** (FIG. **50**), a secondary detent component **236** (FIG. **51**), and a locking lug component **274** (FIG. **49**). As the cable pull actuator **224** pulls on the cable pull arm **234** of the actuator **220** to pivot the camera wing **213** from the flush position to the extended position, the rotation of the cable pull arm **234** is translated to rotation of the camera wing **213** relative to the bracket **228** fixed to the vehicle at which the camera wing assembly **213** is attached.

[0104] As shown in FIG. **45B**, the locking lug **274** engages and is fixed rotationally to the cable pull arm **234** and also engages the bracket **228** through the cable pull arm **234** to, when in a lowered position, fix the cable pull arm **234** relative to the bracket **228** and prevent the cable pull arm **234** (and therefore camera wing) from rotating. The locking lug **274** component is engaged with the bracket **228** when the camera wing **213** is in the flush position until the cable **226** is pulled. The locking lug **274** is connected to the cable pull arm **234** via a screw interface and the locking lug **274** raises or lifts out of engagement with the bracket **228** as the arm **234** is rotated via the cable pull actuator **224** (i.e., the cable pulls the cable pull arm), compressing the locking lug compression spring **276** that pushes the locking lug **274** into engagement with the bracket **228**. When the locking lug **274** is lifted, the detent ring **278** is able to rotate, with the wing **213** connected to the detent ring **278** via the primary detent **238**.

[0105] The locking lug **274** prevents the lower portion of the camera wing pivot (the portion from which the cable pull arm extends) from rotating when the camera wing **213** is manually rotated. The lower portion of the camera wing pivot can only rotate if the cable **226** is pulled. Because the locking lug **274** provides connection between the cable pull arm **234** and the bracket **228**, the locking lug holds the cable pull arm **234** in place when the camera wing **213** is manually extended and ensures that the wing rotates to the correct angle. In other words, the locking lug **274** prevents creation of slack in the cable **226** when the wing **213** is manually extended by resisting the outside force manually rotating the camera wing **213** from acting on the cable **226** and cable pull actuator **224**, which could result in the cable **226** being pushed toward the cable pull actuator **224** and thus bent, damaged, or disconnected. Thus, when the camera wing **213** is manually pivoted, the locking lug **274** engages the cable pull arm **238** and the bracket **228** to fix the cable pull arm **234** relative to the bracket **228**, preventing the movement of the camera wing **213** from causing movement of the camera pull arm **234**, effectively disengaging the cable pull actuator **224** from the camera wing **213** during such manual pivoting of the camera wing.

[0106] The locking lug **278** also allows the return torsion spring force to be low and thus, a smaller torsion return spring **268** may be used. Use of a smaller torsion return spring **268** decreases the force load on the cable pull actuator **224** and on the cable **226** and also decreases the force loss from the cable pull, which aides in situations where more force than usual is needed to extend the wing from the flush position to the extended position (such as when ice builds up at the exterior surface of the vehicle).

[0107] As shown in FIG. **46**, the locking lug **274** is keyed to the detent ring **278** with the locking lug compression spring **276** disposed between the locking lug and detent ring. As the cable pull arm **234** is pulled, the detent ring **278** is held in place by the locking lug's connection to the wing bracket **228**. The cable pull causes the locking lug **274** to lift to release the locking lug from its engagement with the bracket **228**. Once the locking lug lifts, the detent ring **278** and locking lug

274 are able to rotate. The camera wing **213** is connected to and rotates with the detent ring **278** via the primary detent **238**.

[0108] As shown in FIG. **47**, the primary detent component **238** connects the cable pull arm **234** and the camera wing **213** via the detent ring **278**. The primary detent **238** separates or disengages from the camera wing **213** during manual extension or a forward fold (when the camera wing is rotated beyond the extended position such as upon impact with an object). When the primary detent **238** is engaged, it lifts the secondary detent **236** out of position. Thus, there is no position in which more than one of the primary or secondary detent is engaged, providing more control and more consistent detent forces in every position. The primary detent **238** is loaded with an internal compression spring **252** to provide necessary retention forces to retain the camera wing **213** in the detent positions.

[0109] As shown in FIGS. **45A** and **48**, the secondary detent **236** is disposed on top of the camera wing **213** and connects the camera wing **213** with an upper portion of the bracket **228** and only makes connection with the bracket **228** when the camera wing **213** is manually extended (i.e., manually rotated from the park position to the extended position). The secondary detent **236** may only engage the camera wing **213** with the bracket **228** once the camera wing **213** has reached the extended position (such as about 70 percent of the camera wing's total rotational spectrum). In other words, when the camera wing **213** is manually extended, the secondary detent **236** provides a stop point for the camera wing **213** and provides a detent force to retain the camera wing **213** in the extended position. The detent force of the secondary detent **236** is determined by the ramp angle of the detent surfaces in addition to the secondary detent compression spring positioned above the secondary detent **236** and retained by the secondary detent cap.

[0110] Additionally, the secondary detent **236** holds the camera wing **213** in place during an electric resync, allowing electric recovery from the manually extended position. In other words, when the wing **213** has been manually extended, the secondary detent **236** engages the camera wing **213** and the upper portion of the bracket **228** to hold the camera wing **213** in place and the cable pull actuator **224** can be activated to pull on the cable **226**. The secondary detent **236** prevents the wing **213** from folding forward beyond the extended position as the cable is pulled and allows the primary detent **238** to re-engage. When the primary detent **238** re-engages, the secondary detent **236** is forced to release. Once the primary detent **238** is engaged, the wing **213** can be deployed and retracted via operation of the cable pull actuator **224**, as normal.

[0111] Thus, and as shown by the flowchart **1000** in FIG. **52**, the camera wing is pivotable between the flush or retracted position and the extended position manually or via electric activation of the cable pull actuator. When in the retracted position, the primary detent is engaged with the camera wing. When electrically extended, the primary detent remains engaged and the camera wing is pivoted to the extended position with the primary detent engaged. If electrically retracted from extended position with the primary detent engaged, the primary detent remains engaged and the camera wing is pivoted to the retracted position. Thus, the primary detent does not release when the camera wing is operating electrically.

[0112] If, with the primary detent engaged at the extended position, the camera wing is retracted manually, the camera wing is retracted with both primary and secondary detents disengaged (a no-detent position). From the retracted, no-detent position, one activation of the electric cable pull actuator returns the cable pull arm into position via the torsion spring and brings the primary detent back into engagement and the camera wing remains retracted.

[0113] When the camera wing is extended manually, the primary detent is disengaged and the secondary detent is engaged. If the camera wing, in the extended position with the secondary detent engaged, is retracted manually, the primary detent re-engages and the camera wing is pivoted to the retracted position. If the camera wing, in the extended position with the secondary detent engaged, is retracted electrically, one electric activation of the cable pull actuator engages the primary detent in the extended position and a second electric activation of the cable pull actuator pivots the camera

wing to the retracted position with the primary detent engaged.

[0114] If, when in the extended position (either with the primary or secondary detent engaged), the camera wing is manually pivoted beyond the extended position in a direction opposite the retracted position, the camera wing **213** will fold forward toward a forward fold position and then return to its original extended position via the forward fold flap assembly **280**.

[0115] As shown in FIG. **54**, to enable the camera wing **213** to pivot beyond the extended position, there is a space or forward cavity **290** formed in the outer surface of the vehicle that at least partially receives the camera wing **213** when the wing is pivoted toward the forward fold position. The forward fold flap assembly **280** seals the forward cavity **290** when the camera wing **213** is pivoted between the retracted and extended positions and provides a biasing force that, when the camera wing **213** is pivoted beyond the extended position, urges the camera wing **213** toward the extended position. The forward fold flap assembly **280** includes a flap **282** that covers or fills the forward cavity **290** in the vehicle sheet metal that accommodates the forward rotation of the camera wing **213**. A compression spring **284** and linkage **286** provide the force necessary to keep the flap **282** closed over the cavity **290** as well as bias the camera wing **213** back toward the extended position after the wing is rotated beyond the extended position. The forward fold flap assembly **280** includes a pivot element **288** that may be integrally formed with the flap **282**, the pivot element **288** connecting the linkage **286** to the flap **282**, whereby the flap **282** rotates responsive to lateral movement of the linkage and spring **284**. The pivot element **288** pivots about the same pivot axis as the camera wing **213** and assembles to the rest of the camera wing **213** to form a singular assembly that can be installed into the bracket **228**.

[0116] Because the forward fold flap assembly **280** only engages the camera wing **213** via engagement of the flap **282** at the wing **213**, the forward fold flap assembly **280** only provides a biasing force on the wing when the wing is manually pivoted beyond the extended position toward the forward fold position. Additionally, because the cable pull actuator **224** is effectively disengaged from the rotation of the wing when the wing is manually pivoted (via the locking lug **274** engaging the cable pull arm **234** and bracket **228** to fix the cable pull arm **234** relative to the bracket **228**), the forward fold flap assembly **280** may provide the only biasing force on the wing to return the wing from the forward fold position toward the extended position. In other words, pivoting of the camera wing **213** via the cable pull actuator **224** between the retracted and extended positions is performed independent of pivoting the camera wing from beyond the extended position toward the extended position.

[0117] For example and as shown in FIGS. **57A-59B**, the cable pull actuator **224** may pull the cable **226** to pivot the cable pull arm **234** toward the actuator and thus pivot the wing **213** from the retracted position (FIGS. **57A** and **57B**) to the extended position (FIGS. **58A** and **58B**). When the cable pull actuator **224** is operated to pivot the wing **213** from the extended position to the retracted position, the actuator may release the cable **226** or allow for the cable to move in the opposite direction, whereby the torsion spring **268** biases the wing toward the retracted position. The forward fold flap assembly **280** remains stationary (i.e., with the forward flap **282** in the sealed position) when the wing **213** is pivoted between the retracted and the extended positions. As shown in FIGS. **57A** and **58A**, the cable pull arm **234** of the wing pivot pivots when the wing **213** is pivoted between the retracted and extended position while the linkage **286** and pivot element **288** remain stationary.

[0118] When the wing **213** is manually pivoted beyond the extended position toward the forward fold position (FIGS. **59A** and **59B**), the locking lug **236** may engage the cable pull arm **234** and fix the cable pull arm **234** relative to the bracket **228** so that the cable pull arm, the cable, and the cable pull actuator are not subject to the force of the wing **213** manually pivoting toward the forward fold position. Thus, when the wing **213** is pivoted beyond the extended position toward the forward fold position, the cable pull actuator **224** and torsion spring **268** are effectively disengaged from the pivotable movement of the wing via the locking lug **274** fixing the cable pull arm **234** to the

bracket **228**. As shown in FIG. **59A**, movement of the wing **213** toward the forward fold position pushes the flap **282** into the forward cavity **290** and the compressed forward return spring **284**, via the forward return linkage **286** and flap **282**, provides the biasing force to return the wing **213** to the extended position when the wing is released (i.e., when the force that moved it forward is removed). Additionally, because the forward return spring **284** may provide the biasing force on the wing **213** via engagement of the forward fold flap **282** and the wing, the forward return spring **284** only provides the biasing force when the wing **213** engages the forward fold flap **282** and pushes the forward fold flap **282** into the forward cavity **290**.

[0119] Thus, when the camera wing **213** is manually pivoted beyond the extended position, the camera wing **213** may push against the flap **282** which, via the pivot element **288** and linkage **286**, compresses the forward fold return spring **284**. The primary detent **238** disengages when the wing **213** is manually rotated and the rotation of the camera wing **213** forward pushes or slides the flap **282** into the cavity **290** in the side of the vehicle. The camera wing **213** occupies at least a portion of the cavity **290** as it is rotated forward. After the impact, the return spring **284**, via the forward return linkage **286**, pushes the camera wing **213** back to the extended position and returns the flap **282** over or in the cavity **290**. The compression spring **252** of the primary detent **238** pushes the primary detent **238** back into engagement with the camera wing as the camera wing **213** returns to the extended position following the impact.

[0120] Changes and modifications in the specifically described embodiments may be carried out without departing from the principles of the present invention, which is intended to be limited only by the scope of the appended claims as interpreted according to the principles of patent law.

Claims

1. A vehicular camera assembly, the vehicular camera assembly comprising: a mounting base configured to mount the vehicular camera assembly at a side of a vehicle equipped with the vehicular camera assembly; an arm pivotally mounted at the mounting base and pivotable relative to the mounting base; a camera disposed at an outer end region of the arm; wherein, with the vehicular camera assembly mounted at the vehicle, the arm is pivotable relative to the mounting base between at least (i) a drive position, where the camera is positioned outward from the side of the vehicle so that the camera is positioned for viewing rearward of the vehicle, and (ii) a folded position, where the camera is positioned inward from the drive position toward the side of the vehicle; an actuator operable to pivot the arm between the folded position and the drive position; wherein, with the vehicular camera assembly mounted at the vehicle, and when the arm is pivoted from the drive position toward a forward fold position, the arm is at least partially received in a forward cavity at the side of the vehicle; and a forward flap assembly that seals the forward cavity when the arm is pivoted toward the forward fold position, and wherein the forward flap assembly, when the arm is pivoted away from the folded position and beyond the drive position toward the forward fold position, engages the arm and biases the arm toward the drive position.
2. The vehicular camera assembly of claim 1, wherein the actuator is operable to pivot a detent assembly to pivot the arm between the folded position and the drive position.
3. The vehicular camera assembly of claim 2, wherein the detent assembly comprises a (i) primary detent that is engaged when the arm is pivoted between the folded position and the drive position via operation of the actuator, (ii) a secondary detent that is engaged to maintain the arm at the drive position when the arm is manually pivoted to the drive position, and (iii) a locking component that is engaged when the arm is manually pivoted between the folded position and the drive position.
4. The vehicular camera assembly of claim 3, wherein the detent assembly comprises an internal biasing element that imparts a biasing force at the primary detent to urge the primary detent into engagement.
5. The vehicular camera assembly of claim 3, wherein, when the arm is manually pivoted between

the folded position and the drive position, the primary detent is not engaged.

6. The vehicular camera assembly of claim 5, wherein, when the arm is manually pivoted to a position between the folded position and the drive position, the secondary detent is not engaged.

7. The vehicular camera assembly of claim 2, wherein the actuator comprises a cable drive actuator that is positioned remote from the detent assembly and is operable to pivot the detent assembly by pulling a flexible cable connected at the cable drive actuator and the detent assembly.

8. The vehicular camera assembly of claim 7, wherein pulling the flexible cable pivots a cable pull arm attached to the detent assembly to pivot the arm.

9. The vehicular camera assembly of claim 8, wherein the detent assembly comprises a locking component that, when engaged, engages the cable pull arm and the mounting base to fix the cable pull arm to the mounting base when the arm is manually pivoted between the folded position and the drive position.

10. The vehicular camera assembly of claim 8, wherein, when the cable drive actuator is operated to pivot the arm toward the drive position, the cable drive actuator pulls the flexible cable to pivot the cable pull arm in a first direction, and wherein, when the cable drive actuator is operated to pivot the arm toward the folded position, the cable drive actuator releases the flexible cable to pivot the cable pull arm in a second direction opposite the first direction.

11. The vehicular camera assembly of claim 10, wherein the cable drive actuator comprises a torsion spring that biases the cable pull arm in the second direction.

12. The vehicular camera assembly of claim 1, wherein, with the vehicular camera assembly mounted at the vehicle, and with the arm in the folded position, the arm is at least partially received in a recess at the side of the vehicle.

13. The vehicular camera assembly of claim 1, wherein, with the vehicular camera assembly mounted at the vehicle, the arm is manually to a forward fold position, where the arm and camera are pivoted beyond the drive position in an opposite direction from the folded position.

14. The vehicular camera assembly of claim 1, comprising an electronic control unit (ECU) having electronic circuitry and associated software, wherein, with the vehicular camera assembly mounted at the vehicle, the camera, when the arm is in the drive position, captures video image data and the captured video image data is provided to the ECU, and wherein, with the vehicular camera assembly mounted at the vehicle, video image data captured by the camera and provided to the ECU is processed at the ECU for at least one selected from the group consisting of (i) display of video images derived from video image data captured by the camera and provided to the ECU and (ii) detection of an object viewed by the camera.

15. A vehicular camera assembly, the vehicular camera assembly comprising: a mounting base configured to mount the vehicular camera assembly at a side of a vehicle equipped with the vehicular camera assembly; an arm pivotally mounted at the mounting base and pivotable relative to the mounting base; a camera disposed at an outer end region of the arm; wherein, with the vehicular camera assembly mounted at the vehicle, the arm is pivotable relative to the mounting base between at least (i) a drive position, where the camera is positioned outward from the side of the vehicle so that the camera is positioned for viewing rearward of the vehicle, and (ii) a folded position, where the camera is positioned inward from the drive position toward the side of the vehicle; an actuator that is electrically operable to pivot the arm between the drive position and the folded position; wherein the actuator comprises a primary detent interface that is engaged to maintain the arm at the drive position when the arm is pivoted to the drive position; wherein the actuator comprises a secondary detent interface; wherein the arm is manually pivotable relative to the mounting base, and wherein, when the arm is manually pivoted between the folded position and the drive position, the secondary detent interface is disengaged; wherein, when the arm is pivoted between the folded position and the drive position via the actuator, the secondary detent interface is engaged and the primary detent interface is disengaged; wherein the actuator comprises an output gear disposed at a pivot post structure, and wherein a biasing element (i) urges the output gear

along the pivot post structure toward the mounting base and (ii) urges a middle detent element along the pivot post structure into engagement with a lower detent surface of the pivot post structure; a forward flap assembly that seals a forward cavity when the arm is between the folded position and the drive position, and wherein the forward flap assembly, when the arm is pivoted from the drive position toward a forward fold position, engages the arm and biases the arm toward the drive position; wherein, with the vehicular camera assembly mounted at the vehicle, and with the arm in the drive position, the camera captures video image data; and wherein, with the vehicular camera assembly mounted at the vehicle, video image data captured by the camera is processed by an image processor for at least one selected from the group consisting of (i) display of video images derived from video image data captured by the camera and (ii) detection of an object viewed by the camera.

16. The vehicular camera assembly of claim 15, wherein (i) a base portion of the pivot post structure comprises an upper detent surface, (ii) a lower housing portion comprises a lower housing detent surface, (iii) the middle detent element comprises an upper detent surface, (iv) an upper detent element comprises the lower detent surface of the pivot post structure, and (v) the biasing element is disposed between the output gear and the middle detent element.

17. The vehicular camera assembly of claim 16, wherein the lower housing portion is disposed along the pivot post structure, and wherein the lower detent surface of the lower housing portion engages the upper detent surface of the base portion of the pivot post structure.

18. The vehicular camera assembly of claim 17, wherein the output gear is rotatably disposed along the pivot post structure, and wherein ramp portions of the output gear engage lift ramps of the base portion of the pivot post structure, and wherein the output gear is movable between a raised position and a lowered position along the pivot post structure as the ramp portions engage and travel along the lift ramps.

19. The vehicular camera assembly of claim 18, wherein the middle detent element is disposed along the pivot post structure, and wherein at least a portion of the middle detent element is disposed within a cavity of the output gear.

20. The vehicular camera assembly of claim 19, wherein the middle detent element is rotatably fixed relative to the output gear.

21. The vehicular camera assembly of claim 19, wherein the biasing element is disposed entirely within the cavity of the output gear.

22. The vehicular camera assembly of claim 21, wherein the upper detent element receives the pivot post structure and is rotatably fixed relative to the pivot post structure.

23. The vehicular camera assembly of claim 22, wherein the middle detent element, the biasing element, the output gear, and the lower housing portion are disposed along the pivot post structure between the lower detent surface of the upper detent element and the base portion of the pivot post structure.

24. The vehicular camera assembly of claim 16, wherein the primary detent interface comprises part of the upper detent surface of the base portion and a correspondingly formed part of the lower detent surface of the lower housing portion, and wherein the secondary detent interface comprises part of the upper detent surface of the middle detent element and a correspondingly formed part of the lower detent surface of the pivot post structure.

25. The vehicular camera assembly of claim 16, wherein the biasing element urges the middle detent element into engagement with the upper detent element.

26. The vehicular camera assembly of claim 15, wherein the biasing element comprises a wave spring.
